

Assignment 2

SPSS For Research

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Deadline: 15 August 2026

Submission: A report in PDF form. Please write your name as per the certificate. Include the Syntax at the appendix of the report.

Problem Statement:

Nexus Solutions Ltd. is a technology services company with roughly 1,900 employees spread across urban, suburban and rural offices. There is an upcoming board meeting to be held where a few decisions are to be made:

1. Whether to pay the employees to take formal training related to their responsibilities, so that the employees can become more productive and earn more. If evidence shows that having formal training increases employees' pay then the company will invest on the employees.
2. The board also want to know whether employees with more experience and formal education are earning more in the company.

You have joined the HR Analytics unit as a data analyst. Your line manager has extracted an anonymised random sample of 250 employee records and asked you to produce an evidence brief for the Board.

Import salary regression data.csv into SPSS. Each row is one employee. The extract has already been anonymised. No further cleaning has been done for you.

Variable	Description	Unit	Level
employee_id	Anonymised employee identifier	—	Nominal
age	Age at time of extract	Years	Scale
education_years	Total years of completed formal education	Years	Scale
experience_years	Total years of professional work experience	Years	Scale
training_hours	Company-funded training received in the last 12 months	Hours	Scale
region	Office location type	Urban / Suburban / Rural	Nominal
annual_salary_k	Gross annual salary	USD thousands	Scale

Before you analyse anything: check that all 250 cases have imported, set the measurement level and value labels correctly for every variable, and screen each variable for missing values and impossible values. Report anything you find (or state that you found nothing) in one short paragraph.

Task 1:

Produce a table of descriptive statistics for the five scale variables: age, education_years, experience_years, training_hours and annual_salary_k. Your table must report, for each variable:

Complete cases (n), minimum, maximum, mean, median, and standard deviation

Hint:

Analyze > Descriptive Statistics > Frequencies

Analyze > Descriptive Statistics > Explore also works

Analyze > Tables > Custom Tables gives you the most control over layout

Task 2:

Choose and produce graphs appropriate to the variables. You must submit at least the following four, each with proper axis labels.

	Graph
2a	Histogram of annual_salary_k
2b	Bar chart or pie chart of region
2c	Boxplot of annual_salary_k split by region
2d	Scatterplots of annual_salary_k against education_years, experience_years and training_hours, each with a fitted line

Hint:

Graphs > Chart Builder is the flexible route

Graphs > Legacy Dialogs is faster for a single standard chart

Task 3:

Run a Pearson correlation analysis on all five scale variables and present the results as one correlation matrix table that should look like the following (values are masked with ### here):

Table 2: Pearson Correlation

	1	2	3	4	5
1. Age	-				
2. Education Years	###	-			
3. Experience Years	###	###	-		
4. Training Hours	###	###	###	-	
5. Annual Salary	###	###	###	###	-

****.** Correlation is significant at the 0.01 level (2-tailed).

***** Correlation is significant at the 0.05 level (2-tailed).

Hint:

Analyze > Correlate > Bivariate. In options keep Pearson, keep two-tailed, and tick “Flag significant correlations”.

Edit the table in Microsoft Word to make it suitable for APA formatting.

Task 4: Now run a linear regression model to answer “To what extent do years of formal education, years of professional work experience and hours of company-funded training predict annual salary among Nexus Solutions employees?”

1. How much of the variation in salary can these three factors together account for?
2. What is the estimated change in annual salary associated with one additional year of education, holding experience and training constant? And for one additional year of experience?
3. Is the money currently spent on training associated with measurably higher pay?

The regression table should be constructed as follows (values are masked with ### here):

Table 3: Linear Regression Model for the Prediction of Annual Salary

					95% CI	
	b	SE	t	p	LB	UB
Constant	###	###	###	<.001	###	###
Education Years	###	###	###	<.001	###	###
Experience Years	###	###	###	<.001	###	###
Training Hours	###	###	###	.010	###	###

Hint:

Analyze > Regression > Linear